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An Improved EVB Model for Proton Transport in Polymer Electrolyte Membrane

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We have performed a detailed analysis of proton solvation and transport properties in hydrated Nafion using molecular dynamics simulation. The revised EVB model predicts a significantly enhanced transport in comparison with the classical hydronium models, which largely improves the agreement with the available experimental data. The excess proton was found to have weaker interaction with the sulfonate groups in comparison with the classical hydronium ion, reflecting the delocalization nature of the proton charge defects in the EVB model. Our results suggest that the hopping transport has a large contribution to the proton diffusion even at lower hydration levels, as low as 3. It was also found that the Grothuss component becomes more important and its contribution increases linearly as the hydration level increases.

Introduction

Polymer electrolyte fuel cells (PEFCs) have been developed as high energy efficiency and purity energy sources for transportation and stationary power applications. The heart of PEFCs is polymer electrolyte membrane (PEM) that separates the reactant gases and conducts protons. Perfluorosulfonic acid (PFSA) membranes, such as Nafion developed by DuPont, are the most widely used for PEM. It is generally known that the efficiency of proton transport (PT) in PEM is one of the dominant factors controlling power generating efficiency. In the previous research, many macroscopic simulations based on continuum theory have found it difficult to discuss the structural and dynamic properties of proton because PT is largely attributed to the morphology of membrane as well as the surrounding environment of water molecules in nanoscale (1). Thus, attention is focused on analysis of the fundamental process of proton transfer in the confined water environment of hydrated Nafion at nanoscopic level.

In the bulk aqueous solution, it is observed that proton mobility is roughly 5 times higher than other cations such as K^+ which have an ionic radius similar to the hydronium ions. This property of PT is attributed to a combination of proton hopping, known as the Grothuss mechanism (2), and vehicular mechanism where protons diffuse in solution as a hydronium ion, which is the smallest stable hydrated form of the excess proton in water. In Nafion membranes, it is considered that proton hopping transport still takes place although the water dynamic property in the bulk aqueous solution is inhibited by a lack of bulklike water structure at low water contents. PT properties including hopping mechanism in hydrated polymer membrane have been studied using the quantum hopping (Q-HOP) MD method (3, 4) and the MS-EVB method (5). In their studies, there is a restriction of hopping of a single proton among water molecules while the remaining protons residing on hydronium ions were not allowed to hop during the simulation. This

limitation can have an influence on proton dynamic especially at low hydration levels. The self-consistent multistate empirical valence bond (SCI-MS-EVB) methodology (6, 7) has also been used to capture the phenomena of multiproton systems, giving very deep insight and valuable results of proton mobility in hydrated Nafion. However, because the MS-EVB method requires 8-10 EVB states per proton in combination with several water molecules as large as 4, it prohibits the efficient application of the model to investigate the influence of proton hops under confined environments at very low hydration levels. In this study, using a revised EVB model (8) based on the previous study of the two-state EVB model reported by Walbran et al. (9), we have investigated the influence of the proton hopping transport on the structure and dynamic properties of protons in hydrated Nafion membrane at different hydration levels.

Simulation Methods

Nafion chains with equivalent molecular weight ~ 1100 have been employed in the simulation. We generated a Nafion chain by linking the polar monomeric unit (P) to the end of the nonpolar monomeric unit N_7 , repeating the procedure 10 times to obtain $(N_7P)_{10}$. The potential model for Nafion was based on the DREIDING force field as referred to in previous studies by other authors (10-13). The aSPC/Fw model (14) for both hydronium ions and water molecules are employed. To capture the underlying physics of multiproton systems, the EVB approach (8) has been used in this work. In the EVB model, a simplified cluster model of the Zundel cation, where a hydronium ion exchanges a proton with a neighboring water molecule, were considered. All the parameters for the EVB model were taken from the previous study (8).

The initial configurations were generated by randomly placing 4 Nafion chains in a box with periodic boundary condition in all directions. To ensure charge neutrality, we added a total of 40 hydronium ions and considered water contents of $\lambda = 3, 6$, and 12, where the parameter λ indicates the ratio of the number of water molecules to that of SO_3^- ($N_{H_2O, H_3O^+}/N_{SO_3^-}$). After all molecules were added, the annealing procedure was carried out to establish the equilibrium state. The annealing procedure was performed based on a combination of a number of annealing methods proposed previously (10, 15-18). The simulation cell was firstly compressed gradually to the corresponding experimental density (19) at each water content, and *NPT* MD simulation at 800 K and 100 MPa using the L-J potential parameters, ϵ , reduced to 1/100 magnitude among Nafion atoms (15). Then, the *NPT* ensemble was performed to gradually return the L-J parameters to their original values and the conditions to $T = 300$ K and $P = 1$ atm. After that, MD simulations were performed by varying the temperature five times between 300 K and 800 K for 1 ns, followed by equilibration *NPT* MD simulation at $T = 300$ K and $P = 1$ atm for 5 ns to attain the final density. We used this final structure for the additional *NPT* ensemble for 3 ns to collect the trajectory of molecules every 1.0 ps to analyze the structural and dynamic properties of protons and water molecules in the Nafion membrane. The temperature was held by the Nosé-Hoover thermostat (20), and the pressure was controlled by the Andersen method (21). The smooth particle mesh Ewald method (22) was used to calculate electrostatic interactions.

Results and Discussion

Information on the proton solvation structure around sulfonate groups was obtained by the quantitative analysis of the excess proton distribution using the radial distribution function (RDF) shown as

$$g_{A-B}(r) = \left(\frac{n_B}{4\pi r^2 dr} \right) / \left(\frac{N_B}{V} \right), \quad [1]$$

where $4\pi r^2 dr$ is the volume of a dr thick spherical shell at radius r from particle A, n_B is the number of B particles in the shell, V is the volume of the system, and N_B is the number of B particles in the whole system. The center of excess charge (CEC) defined in the previous studies (6, 7) were used to describe the position of the excess proton. Figure 1 shows the RDFs between the sulfur atom of the sulfonate group and the CEC (S–CEC), and the sulfur atom and the hydronium oxygen atom (S–O_h) at hydration level 6. The peak of S–CEC is broadened in comparison with S–O_h, suggesting the delocalization nature of the proton charge defects. The first minimum was found at ~ 4.4 Å for S–O_h, indicating the sulfonate group gathers water molecules to form the first solvation shell with the radius of ~ 4.4 Å. Similar results were also reported in the previous study (6).

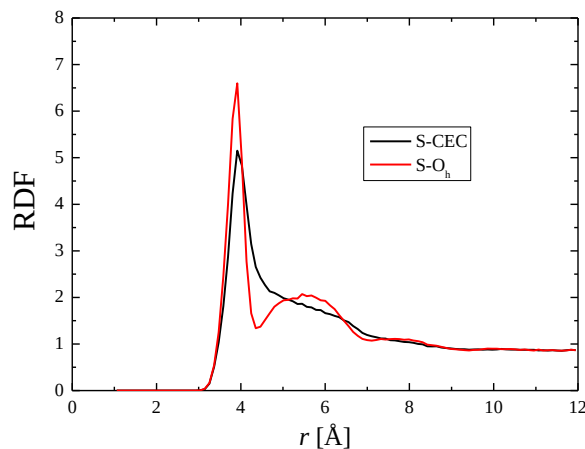


Figure 1. RDFs of sulfur–CEC and sulfur–hydronium oxygen at hydration level 6.

The transport behavior of excess protons and water molecules were examined using the mean square displacements (MSDs) of protons and water molecules. Then, the diffusion coefficients for protons (D_{H^+}) and water (D_w) were determined from the MSDs using the Einstein–Smoluchowski equation written as

$$D = \lim_{t \rightarrow \infty} \frac{1}{6t} \langle |r(t) - r(0)|^2 \rangle, \quad [2]$$

where t is the diffusion time, $r(t)$ is the position vector of particles. Figure 2 (a) shows the MSDs of the CECs as a function of time for the water content of $\lambda = 3, 6$, and 12. The diffusion coefficients of protons were then estimated by the linear regime of these MSD curves from 500 to 1500 ps at each hydration level, and the results are shown in Figure 2 (b). Experimental measurements (23) are also plotted in Figure 2 (b). The diffusion coefficients are in good agreement with the experiment, which show significantly enhanced transport in comparison with the classical hydronium model (24). Figure 3 (a)

shows the MSDs of water molecules as a function of time for various water contents. Using the same calculation approach for the diffusion coefficient of protons, that of water molecules were calculated and compared with the available experimental values obtained from experimental measurements of nuclear magnetic resonance (23, 25) and quasi-elastic neutron scattering (26) in Figure 3 (b). The simulation results are in excellent agreement with experimental data for the studied hydration levels, validating our simulation models.

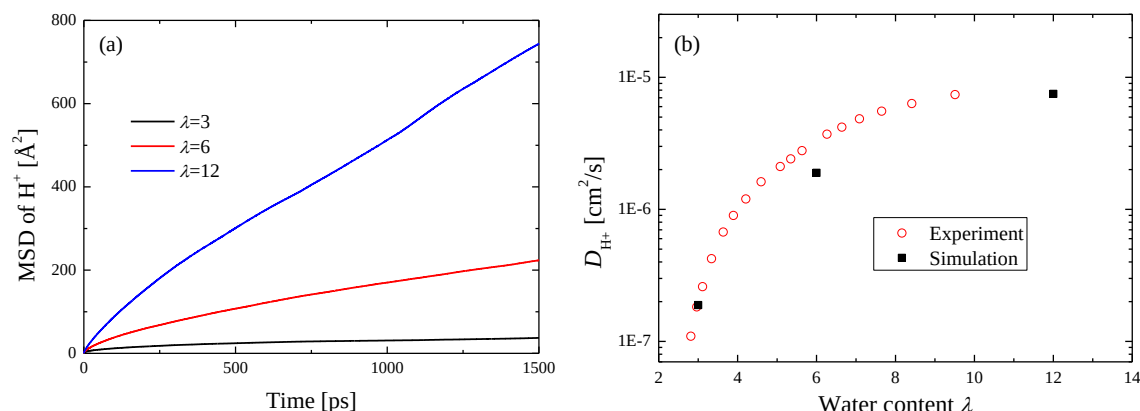


Figure 2. (a) MSDs of CECs at hydration level of 3, 6, and 12. (b) Diffusion coefficients of CECs at each water content.

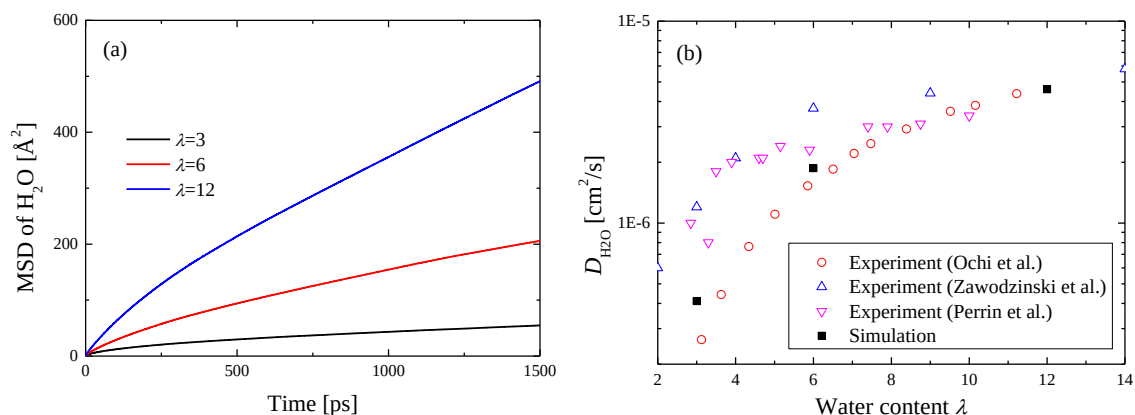


Figure 3. (a) MSDs of water molecules at hydration level of 3, 6, and 12. (b) Diffusion coefficients of water molecules at each water content.

Because the PT is a combination of vehicular and hopping motions, the PT was decomposed into the vehicular and the Grotthuss components to investigate their contributions using the same approach reported in previous study (7). Figure 4 (a) shows the MSDs for both the Grotthuss components (solid lines) and the vehicular components (dashed lines) at different hydration levels. The diffusion coefficients of each component, estimated from the linear regime of the corresponding MSDs, were shown in Figure 4 (b). The results show that the diffusion coefficient of the Grotthuss component, D_G , increase larger than that of the vehicular component (D_V) as increasing hydration level, suggesting the Grotthuss mechanism becomes more important at higher hydration levels. Interestingly, although one might expect that the vehicular transport is dominant due to the small number of hydrogen bond network at a low hydration level of 3, the Grotthuss

contribution, calculated from the ratio of D_G to a sum of D_G and D_V , indicates more than ~50 % contribution to the total PT. In addition, the Grotthuss contribution was found to increase almost linearly with increasing hydration level, reasonably approaching to the value of ~80 % in the bulk aqueous solution (27).

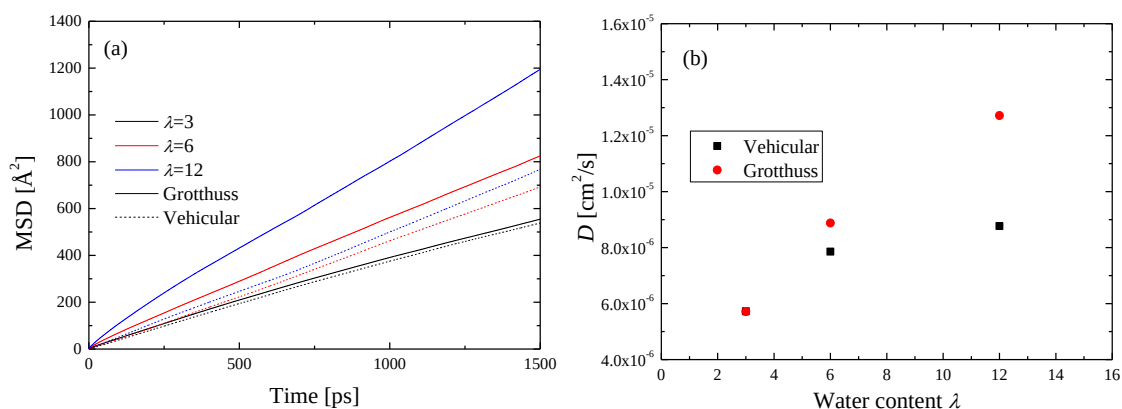


Figure 4. (a) MSDs of vehicular and Grotthuss components at hydration level of 3, 6, and 12. (b) Diffusion coefficients of vehicular and Grotthuss components at each water content.

Conclusions

We have analyzed the structure and dynamic of solvent molecules in hydrated Nafion at different hydration levels using a revised EVB model to investigate the proton solvation and transport properties. The revised EVB model successfully described PT through the Grotthuss hopping mechanism, giving significantly enhanced transport in comparison with the classical hydronium diffusion. It was found that sulfonate groups strongly interact with hydrated protons and water molecules, forming the solvation shells. The RDF between the excess proton and the sulfonate groups was found to have wider peak in comparison with the RDF between the classical hydronium ion and the sulfonate groups, suggesting that the EVB model reflects the delocalization nature of the proton charge defects. The diffusion coefficients of protons and water molecules were found to be consistent with the available experimental data, testifying the validation of the simulation models. It was also found that a hopping-mechanism has a large contribution to the proton diffusion even at very low hydration. The Grotthuss component becomes more important as the hydration level increases, and its contribution increases linearly approaching to the value in the bulk aqueous solution.

Acknowledgments

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